IoT for Kids

Teacher's Guide & Circuit Handbook

A two-day beginner workshop: how to teach it, and how every component connects to the Arduino.

DAY 01 · DISCOVER
How does that work?

DAY 02 · MAKE
From idea to a tiny invention.

PART A · FOR THE TEACHER

1 Workshop at a glance and teaching method
2 Preparation, kit and safety
3 Setup guide (software, board, libraries)
4 Day 1 plan: Discover
5 Day 2 plan: Make
6 Component overview and pin summary
7 Discussion questions, glossary, troubleshooting

PART B · COMPONENT CIRCUITS

1. Arduino Uno R3 Compatible Board
2. Breadboard & Jumper Wires
3. LED Traffic Light Module 5V
4. RGB LED Module
5. PCB Mounted Passive Buzzer Module
6. 74HC595 3-Digit Display Module
7. 0.96 Inch OLED Display (128x64)
8. TowerPro SG90 Micro Servo Motor (180°)
9. Motor Driver L298N + DC Motor
10. TTP223 Touch Switch Module
11. IR Obstacle Avoidance Sensor Module
12. HC-SR04 Ultrasonic Distance Sensor
13. Heat Flame Sensor
14. Raindrop Detection Sensor Module
15. Soil Moisture Sensor Module
16. DHT11 Humidity & Temperature Sensor
17. DS18B20 Digital Temperature Sensor

PART C · PROJECT GUIDES

Wiring and how it works for the six Day 2 inventions. The code is in the separate Code PDF.

1. Workshop at a glance

Item	Plan
Format	Two full days (Day 01 Discover, Day 02 Make), each from **9:00 am to 4:00 pm**, with tea breaks and a one-hour lunch from 12:30 to 1:30.
Who it is for	Beginners. No coding or electronics knowledge is needed; we start with everyday examples and simple activities.
Groups	Teams of 3 to 4 students sharing one kit and one computer. Rotate roles: **Builder** (wires), **Coder** (sketch), **Tester** (checks results), **Reporter** (explains to class).
Big idea	**Sense → Think → Act.** A sensor senses the world, the Arduino thinks (runs the code), and an output acts (light, sound, movement, display).
Final result	Each team presents a tiny invention that senses something and reacts to it.

What students will be able to do

- Explain IoT in everyday words and give examples (smart bulbs, smart watches, automatic doors).
- Tell inputs (sensors) apart from outputs (actuators).
- Wire a sensor or output to an Arduino safely, following a diagram.
- Upload a ready sketch, read the Serial Monitor and change a value to change the result.
- Explain their project as Sense → Think → Act, and work as a team.



How I will teach it: five principles

- **Start from what they know.** Open with smart things from their daily life, then ask 'what does it sense, what does it do?'.
- **Predict → Build → Test → Explain.** Before every build, students predict what will happen. After the test they explain why it did or did not match.
- **Hands-on first, theory in small pieces.** About 70% of the time students are wiring, testing and changing sketches. Explanations stay under 5 minutes.
- **Ready code, small changes.** Teams start from the sample sketches in Part B. Their coding task is to change a number, a pin or a message, which is enough to feel like a creator.
- **Celebrate mistakes.** A wrong wire is a clue, not a failure. Use the troubleshooting list (section 7) and let teams debug each other's circuits.

2. Preparation, kit and safety

Kit for each team

Each team of 3 to 4 students gets one kit with the 17 parts pictured in Part B: Arduino Uno + USB cable, breadboard, jumper wires, traffic light module, RGB LED module, passive buzzer, 3-digit display, OLED, SG90 servo, L298N driver + DC motor, touch switch, IR sensor, HC-SR04, flame sensor, rain sensor, soil moisture sensor, DHT11 and DS18B20. Add a 4×AA battery holder and a small cup of water with a dropper for the rain sensor.

Teacher checklist (1–2 days before)

- Set up every laptop with the Arduino IDE, the board and the libraries (see section 3, Setup guide).
- Open each sample sketch from Part B and save it in a folder named after the component (e.g. 01_Blink, 05_Buzzer) so teams can open it quickly.
- **Test every component yourself** once. Put aside spare modules and replace any that do not respond.
- Print the circuit page for each component (or show it on a screen). Print the colour code poster: red = 5V, black = GND, other colours = signals.
- Prepare the 'Smart or not?' picture cards (smart watch, automatic door, street light, a plain torch, a fan, etc.).
- Set up tables so that each team has space for the kit and a laptop, plus a spill-safe area for the water cup.

Safety rules to tell students on Day 1

- **Unplug the USB cable before changing any wire,** then check your wiring with a partner before plugging in.
- **5V goes to VCC, GND goes to GND.** Swapped power wires can damage a module. Never touch 5V to GND.
- **No water near the Arduino.** The rain sensor plate is the only part that gets drops, and only with the teacher present.
- **No fire for students.** Only the teacher demonstrates the flame sensor, with a small candle at a safe distance (or uses a torch instead).
- **Batteries and motor driver:** the teacher checks the L298N wiring before power is connected; the heatsink can get warm.
- Keep loud buzzers short, and keep long hair, sleeves and fingers away from spinning motors.

Rules of the lab for good teamwork

- Everyone touches the kit and everyone does a role; roles rotate at every break.
- Ask three teammates before asking the teacher (the 'ask 3 then me' rule).
- Put the parts back in the box labelled; pack-up is part of the activity.

3. Setup guide

Do this once on every laptop before the workshop. It takes about 20 minutes.

Step	What to do
1 Install the software	Download the free **Arduino IDE** from arduino.cc/en/software, choose your system (Windows, macOS or Linux) and install it with the default options.
2 Connect the board	Plug the Arduino into the laptop with the USB cable. A light on the board turns on. If the computer does not recognise a compatible board, install the **CH340 USB driver** and plug it in again.
3 Choose board and port	In the IDE open **Tools > Board > Arduino AVR Boards > Arduino Uno**. Then open **Tools > Port** and pick the port that appears when the board is plugged in (COM3, COM4... on Windows, /dev/cu.... on Mac).
4 Test with Blink	Open **File > Examples > 01.Basics > Blink** and click the **Upload** arrow. The small LED on the board should blink once per second. If it does, the setup is complete.
5 Install the libraries	Open **Sketch > Include Library > Manage Libraries**, search each name in the table below and click Install. Install all of them now, so nothing needs the internet in class.
6 Serial Monitor	Many sketches print numbers. Open **Tools > Serial Monitor** and set the speed (bottom right) to **9600 baud** to see them.
7 Wiring routine	Unplug USB > wire the module using the diagram > check each wire with a partner > plug in USB > upload the sketch.
8 Sharing 5V and GND	The Arduino has only one 5V pin. When a project uses several modules, connect Arduino 5V to the breadboard red (+) rail and Arduino GND to the blue (−) rail, then plug each module's VCC and GND wires into the rails.

Libraries to install

Library to search for	Used by
Adafruit SSD1306	OLED display (component 7)
Adafruit GFX Library	OLED display (installed together with SSD1306)
DHT sensor library (by Adafruit)	DHT11 sensor (component 16)
Adafruit Unified Sensor	Needed by the DHT library
OneWire	DS18B20 sensor (component 17)
DallasTemperature	DS18B20 sensor (component 17)
Servo	Servo motor: already built in, nothing to install

How to use the sketches in this handbook

- All sketches are in the separate **Code PDF** (and as ready-to-open files in Arduino_Sketches.zip). Open a new sketch (File > New), delete everything inside, paste the sketch, and click Upload.
- Sketches that print to the Serial Monitor need it open at 9600 baud.
- Save each sketch with a clear name (for example 12_HC-SR04) so teams can reopen it quickly.

4. Day 1 · DISCOVER: How does that work?

Time	Activity	What the teacher does	Parts used
9:00 – 9:20	**Welcome & 'Smart or not?'**	Introductions and the lab rules. Show picture cards; teams sort them into smart / not smart. Ask: how does the automatic door know you are there?	None
9:20 – 10:00	**What is IoT? Sense → Think → Act**	Teach the three-box model with the body analogy: eyes and ears sense, brain thinks, muscles act. List everyday examples on the board in 3 columns.	Display the kit
10:00 – 10:15	Tea break	Teams get their kits ready and swap roles.	
10:15 – 11:00	**Meet the kit & stay safe**	Pass round the Arduino, breadboard and wires (components 1–2 of Part B). Demo the breadboard strips and the 5V / GND idea. Read the safety rules. Each team connects the board and uploads Blink.	Arduino, breadboard, jumper wires
11:00 – 12:30	**First outputs**	Upload Blink together, then each team wires the traffic light, the RGB LED and the buzzer in turn. Predict → test → change timings, colours and notes.	Traffic light, RGB LED, buzzer
12:30 – 1:30	Lunch		
1:30 – 2:30	**First sensors: touch and IR**	Touch switch controls the LED. IR sensor prints 'Obstacle!' in the Serial Monitor. Ask: which one is the input, which one is the output?	Touch switch, IR sensor
2:30 – 2:45	Break	Reset kits, swap roles.	
2:45 – 3:30	**The magic ruler**	HC-SR04 prints distance in the Serial Monitor. Explain echo with the bat story. Challenge: measure the desk, a book, a hand, then make the traffic light turn red when something is close.	HC-SR04, traffic light
3:30 – 4:00	**Recap & exit ticket**	Card sort: each part is a Sensor, a Brain or an Actuator. Pack up. Preview Day 2 and ask each team to think of a problem to solve.	All

5. Day 2 · MAKE: From idea to a tiny invention

Time	Activity	What the teacher does	Parts used
9:00 – 9:30	**Recap game & today's mission**	Quick quiz: 'Sensor or actuator?', 'What does GND mean?'. Introduce today's mission: build a tiny invention that senses and reacts.	–
9:30 – 10:30	**Sensor stations** (4 × 15 min, teams rotate)	Station A: DHT11 + DS18B20 (compare temperature). B: soil moisture. C: rain sensor. D: flame sensor (teacher-run demo). Teams record the readings on a worksheet.	DHT11, DS18B20, soil, rain, flame
10:30 – 10:45	Tea break	Rotate roles.	
10:45 – 11:30	**Show it & move it**	Demo the OLED and 3-digit display for readings, and the servo and motor for movement. Teams try each one quickly.	OLED, 3-digit display, servo, L298N + motor
11:30 – 12:30	**Plan the invention**	Teams choose a project from the menu below, draw their Sense → Think → Act plan, collect the parts and start wiring.	As chosen
12:30 – 1:30	Lunch		
1:30 – 3:00	**Build time**	Teams build and test their invention. Teacher walks round with the 'ask 3 then me' rule, checking power wires first.	As chosen
3:00 – 3:15	Break & tidy	Teams make their projects ready to present.	
3:15 – 3:50	**Show & share**	Each team gives a 2–3 minute pitch: what it senses, how it thinks, what it does. Others ask one question.	Projects
3:50 – 4:00	**Reflect & what's next**	Ask: what was hardest, what would you add? Point to next steps: a school science fair, a robot car, a weather station. Pack up.	–

Project menu (every team picks one)

Invention	Senses	Thinks	Acts	Parts
Smart dustbin	Hand nearby (HC-SR04)	If distance < 15 cm	Lid opens (servo), closes after 3 s	HC-SR04, SG90
Rain alert	Rain drops	If value < limit	Buzzer + red light	Rain sensor, buzzer, traffic light
Plant helper	Soil wetness	If dry	Red / green LED + message on OLED	Soil sensor, RGB LED, OLED
Fire alarm	Flame (teacher-run test)	If flame detected	Alarm sound + red light	Flame sensor, buzzer, traffic light
Mini weather station	Temperature + humidity	Read every 2 s	Show values on OLED	DHT11, OLED
Fan controller (stretch)	Temperature	If > set value	Motor spins	DS18B20, L298N + motor

6. Component overview and pin summary

Use this table as a quick reference while students wire. Day 1 parts are marked D1, Day 2 parts D2.

#	Component	Role	Arduino pins used	Day
1	Arduino Uno R3 Compatible Board	Brain	–	D1
2	Breadboard & Jumper Wires	Support	5V and GND rails	D1
3	LED Traffic Light Module 5V	Output (acts)	D10, D9, D8, GND	D1
4	RGB LED Module	Output (acts)	D9, D10, D11, GND	D1
5	PCB Mounted Passive Buzzer Module	Output (acts)	D8, GND	D1
6	74HC595 3-Digit Display Module	Output (acts)	5V, GND, D8, D9, D10	D2
7	0.96 Inch OLED Display (128x64)	Output (acts)	GND, 5V, A5 (SCL), A4 (SDA)	D2
8	TowerPro SG90 Micro Servo Motor (180°)	Output (acts)	D9, 5V, GND	D2
9	Motor Driver L298N + DC Motor	Output (acts)	D5, D7, D8, GND + battery	D2
10	TTP223 Touch Switch Module	Input (senses)	5V, GND, D2	D1
11	IR Obstacle Avoidance Sensor Module	Input (senses)	5V, GND, D2	D1
12	HC-SR04 Ultrasonic Distance Sensor	Input (senses)	5V, D9, D10, GND	D1
13	Heat Flame Sensor	Input (senses)	5V, GND, D2	D2
14	Raindrop Detection Sensor Module	Input (senses)	5V, GND, A0, D2	D2
15	Soil Moisture Sensor Module	Input (senses)	5V, GND, A0, D2	D2
16	DHT11 Humidity & Temperature Sensor	Input (senses)	5V, D2, GND	D2
17	DS18B20 Digital Temperature Sensor	Input (senses)	5V, D2, GND	D2

7. Discussion questions, glossary and troubleshooting

Discussion questions to ask during the workshop

- Which of the things you use every day are 'smart'? What does each one sense? What does it do?
- Why does a robot need sensors? What would happen without them?
- Why do we need a motor driver instead of connecting a motor straight to the Arduino?
- Would you trust the flame sensor alone in a real building? What could go wrong?
- How can data from sensors help people (farmers, hospitals, cities)? What about privacy?

Simple assessment

- **Exit tickets:** 'One sensor I used, what it senses, and one thing it could control.'
- **Project rubric (4 points):** works as intended · wired safely and neatly · can explain Sense → Think → Act · teamwork.
- **Observation checklist:** reads a wiring diagram, changes a value in code, tests and fixes a problem.

Glossary for students

Word	Kid-friendly meaning
IoT	Internet of Things: everyday objects that sense and act, often connected to the internet
Sensor	A part that senses something (touch, light, distance, heat)
Actuator	A part that does something (light, sound, movement)
Microcontroller	A tiny computer that runs one program (the Arduino)
Sketch	An Arduino program
Digital / Analog	Digital = ON or OFF. Analog = a range of values (0 to 1023)
PWM	Switching a pin on and off very fast to dim a light or change speed
VCC / GND	VCC = power (+). GND = ground (−)

Troubleshooting

Problem	Check this first
Computer does not see the Arduino	Try another USB cable (some only charge), pick the right Port, install the CH340 driver for compatible boards.
Upload fails	Tools > Board must be Arduino Uno; close the Serial Monitor of another program using the port.
Sensor gives no reading or random numbers	Are VCC and GND correct and firm? Is the signal wire on the pin used in the sketch? Is it a loose jumper?
Nothing lights up	Check LED polarity / pin number; try a different pin; confirm the sketch uploaded.
Servo jitters or the board resets	Weak power. Use a separate battery pack and join the grounds.
Motor does not spin	Check battery charge, the ENA jumper, OUT1/OUT2 wires and that Arduino GND joins the L298N GND.
OLED stays blank	SDA and SCL may be swapped; try address 0x3D; confirm the libraries are installed.
DS18B20 shows −127	DATA wire is loose or on the wrong pin.

Arduino Uno R3 Compatible Board



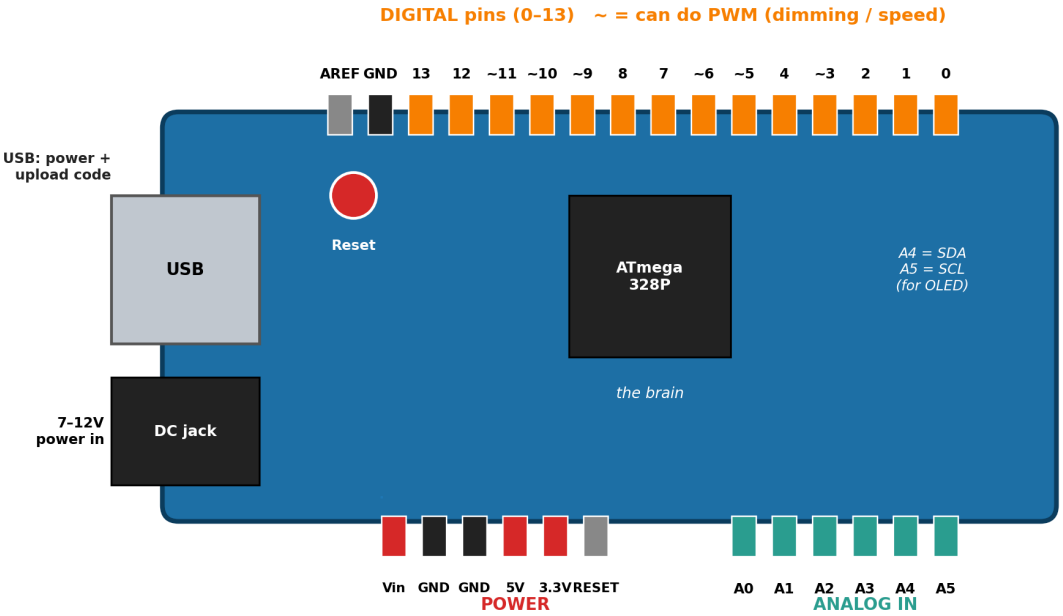
WHAT IT DOES

The main board of every project. It reads the sensors (inputs) and controls lights, motors and displays (outputs).

HOW IT WORKS

A chip on the board (the ATmega328P) follows the instructions, called a sketch, that you upload from the computer through the USB cable. Every pin can be set as an INPUT (to listen) or an OUTPUT (to speak). Digital pins know only ON or OFF; analog pins can read a range of values.

PARTS OF THE BOARD



QUICK REFERENCE

Part	What it does
USB port	Powers the board and uploads your code
Digital pins 0–13	Read buttons/sensors or switch things on and off
~ pins (3,5,6,9,10,11)	PWM: dimming LEDs, motor speed, servo
Analog pins A0–A5	Read values like soil wetness or light
5V and GND	Power for sensors (+ and −)
Reset button	Restarts the program

SETUP STEPS

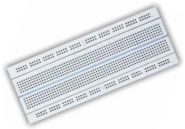
1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Result: The small LED on the board blinks once per second.

WHY IT IS USED

- Smart-home gadgets and quick prototypes
- Robots and school science-fair projects
- Weather stations and data loggers
- Learning to program with real, visible results

In our workshop: It is the brain of the workshop: every sensor and output we use connects to it.

Breadboard & Jumper Wires





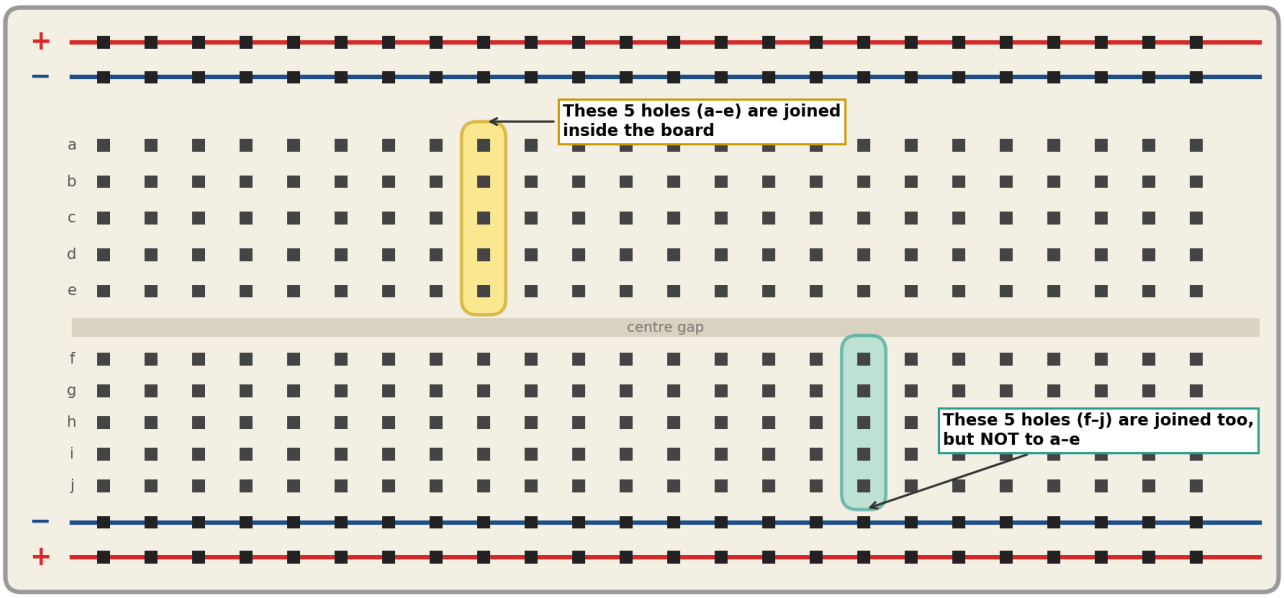
WHAT IT DOES

The breadboard lets students build circuits without soldering. Jumper wires join the Arduino, the breadboard and the modules.

HOW IT WORKS

Under the plastic, metal strips join some holes together. In the middle area each column of 5 holes (a–e, and f–j) is one connection point. The long rails on the sides run the whole length and are used to share 5V (+) and GND (−) with many modules.

HOW A BREADBOARD IS JOINED



Red (+) rail: whole strip is joined, connect 5V here **Blue (−) rail: whole strip is joined, connect GND here**

QUICK REFERENCE

Wire type	Typical use
Male – Male	Arduino header to breadboard, or breadboard to breadboard
Male – Female	Arduino/breadboard pin to a module's pin header
Female – Female	Module pin to module pin
Colour habit	Red = 5V, Black = GND, other colours = signals

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Push the Arduino 5V wire into the red (+) rail and the GND wire into the blue (−) rail.
3. Plug a module's VCC and GND wires into those rails, and its signal wire into a free hole.
4. Check with a partner, then plug in the USB cable.

WHY IT IS USED

- Testing a circuit before making it permanent
- Sharing 5V and GND between many modules
- Moving a wire in seconds when something needs to change
- Building safely and cleanly with no soldering

In our workshop: Students use it to build every circuit and to share power between sensors in the projects.

LED Traffic Light Module 5V



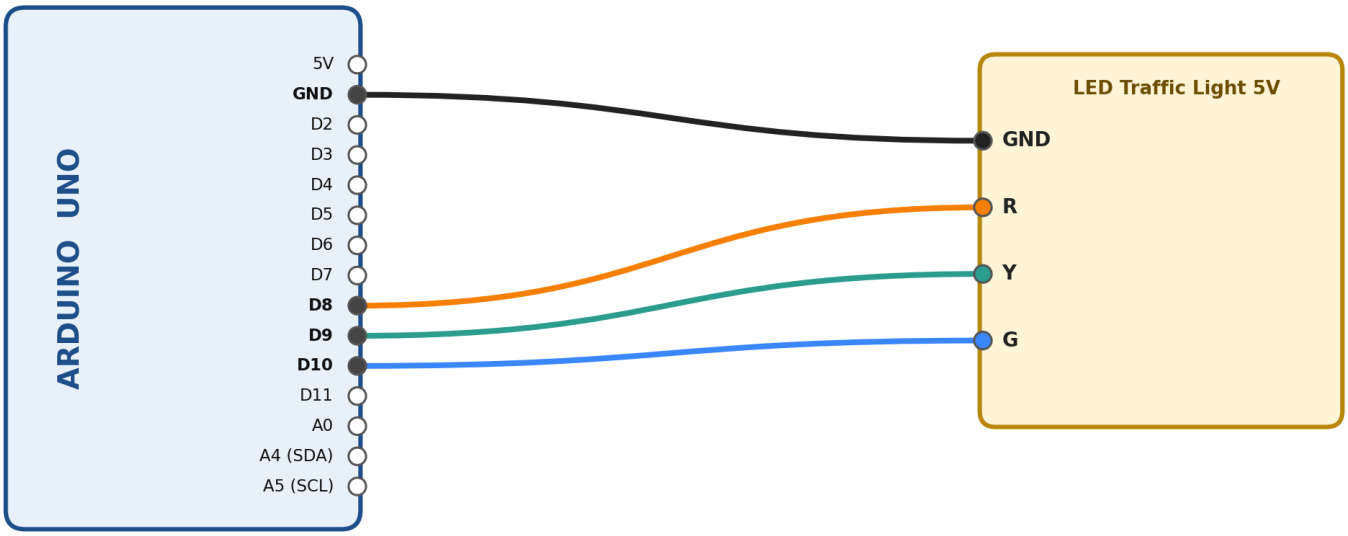
WHAT IT DOES

Red, yellow and green LEDs on one board. Used to model traffic signals and to show status: OK, warning, danger.

HOW IT WORKS

An LED glows when its pin is set HIGH (5V). Small resistors are already on the board, so the module can be connected straight to the Arduino pins.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
G	D10	Green LED
Y	D9	Yellow LED
R	D8	Red LED
GND	GND	Ground

SETUP STEPS

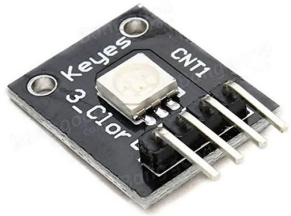
1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Result: Green, yellow and red light up one after another.

WHY IT IS USED

- Traffic-signal models and road-safety lessons
- Status lights on machines (OK, warning, stop)
- Indicators on robots and science projects
- Teaching sequences and timing

In our workshop: A fast first output: students see their code work immediately and learn timing and sequences.

RGB LED Module



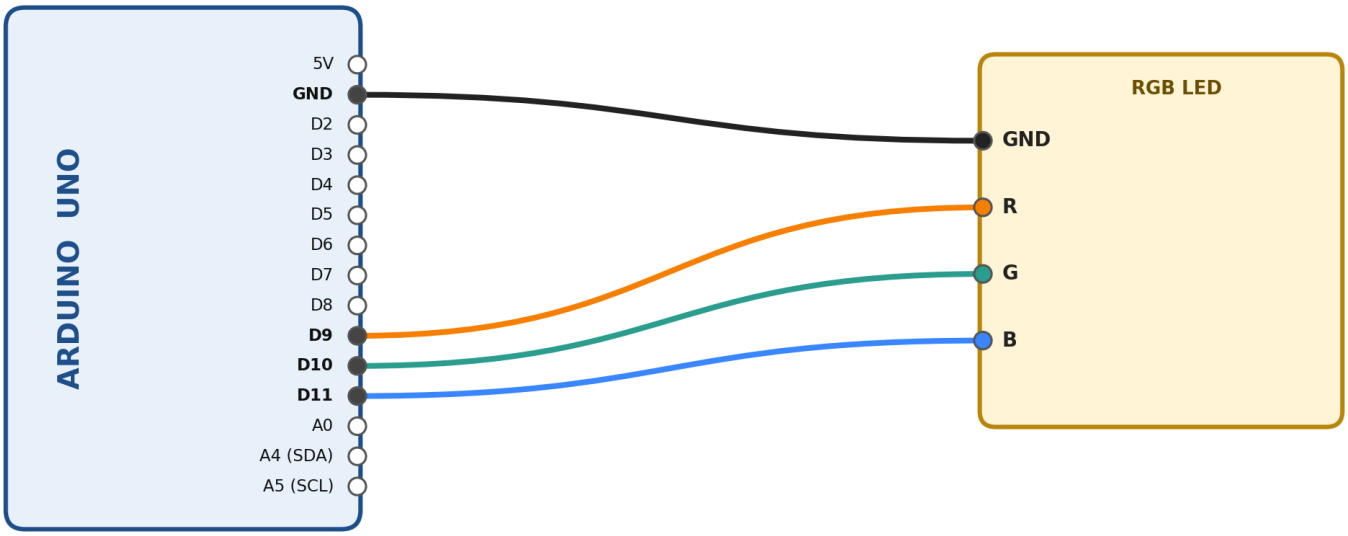
WHAT IT DOES

One LED that can show many colours by mixing red, green and blue. Used for mood lights, colour indicators and status lights.

HOW IT WORKS

Inside are three tiny LEDs. By giving each one a different brightness (using PWM pins and analogWrite, 0–255) we mix colours, exactly like mixing paints but with light.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
R	D9 (~)	Red brightness
G	D10 (~)	Green brightness
B	D11 (~)	Blue brightness
GND (−)	GND	Ground

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Result: The LED shows red, green, blue and yellow in turn.

WHY IT IS USED

- Mood lighting and decorations
- Colour status indicators (red = danger, green = OK)
- Smart lamps and light shows
- Teaching how colours mix

In our workshop: Teaches PWM and colour mixing, and gives a colourful status light for projects like the Plant Helper.

PCB Mounted Passive Buzzer Module



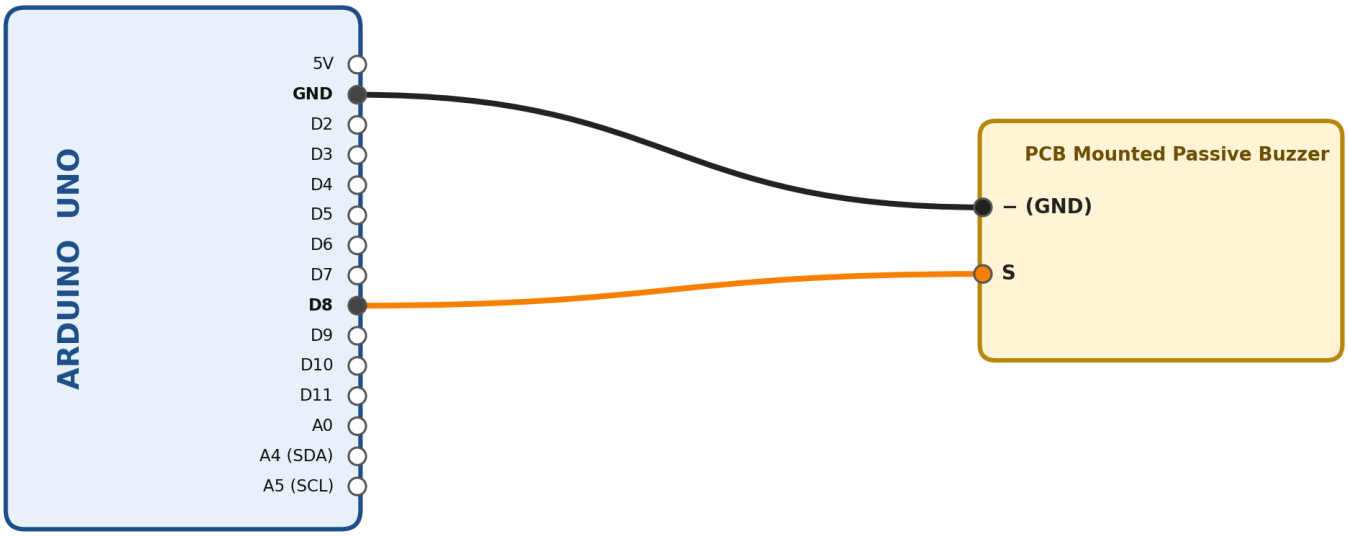
WHAT IT DOES

Makes beeps and musical notes. Used for alarms, alerts and simple melodies.

HOW IT WORKS

A passive buzzer needs a changing signal: the Arduino switches the pin ON and OFF very quickly. How fast it switches (the frequency in Hz) sets the pitch, and the tone() command does this for us.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
S	D8	Signal (tone)
– (GND)	GND	Ground
Middle pin	–	Usually not used. If yours is marked VCC, connect it to 5V

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Result: Five rising notes play, then a short pause.

WHY IT IS USED

- Alarms and alerts (fire, rain, intruder)
- Timers and doorbells
- Beeps for button presses and games
- Playing simple tunes

In our workshop: Gives projects a sound: the rain alert and fire alarm use it to warn people.

74HC595 3-Digit Display Module



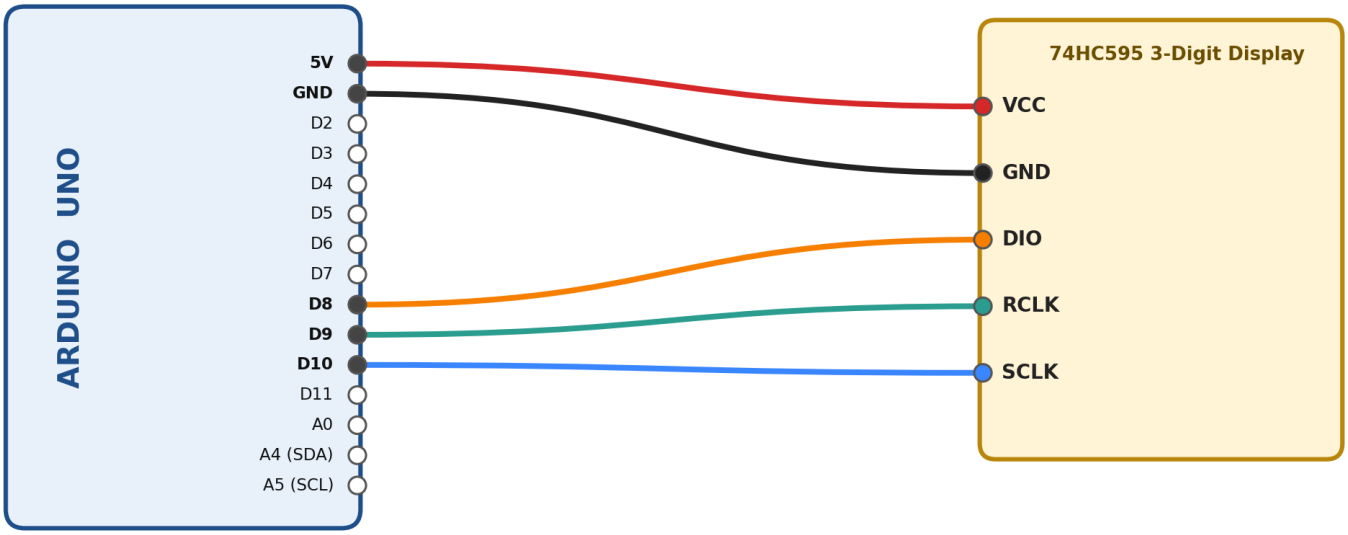
WHAT IT DOES

Three seven-segment digits for showing numbers: counters, scores, temperature or distance.

HOW IT WORKS

A 74HC595 chip lets the Arduino control many segments using only 3 wires. The code sends the pattern for the digit, then which digit to light. By lighting the three digits one after another very quickly, they all look lit together.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
VCC	5V	Power
GND	GND	Ground
DIO	D8	Data in
RCLK	D9	Latch (show it)
SCLK	D10	Clock (shift bits)

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Result: The display shows 123.

WHY IT IS USED

- Digital clocks and timers
- Scoreboards and counters
- Showing temperature or distance
- Price and number displays

In our workshop: Shows numbers on its own, such as counters or a distance reading, without a computer screen.

0.96 Inch OLED Display (128x64)



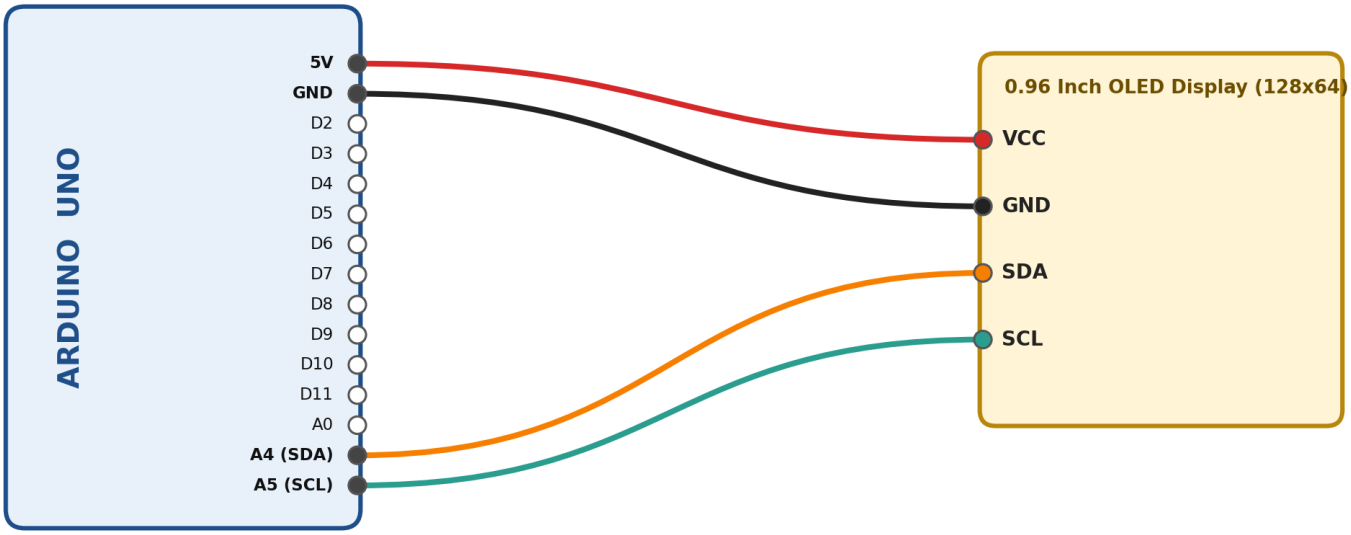
WHAT IT DOES

A tiny screen for text, numbers and simple pictures. Used to show sensor readings and messages without a computer.

HOW IT WORKS

The display talks to the Arduino over I²C, a two-wire connection (SDA = data, SCL = clock). Each pixel makes its own light, so it is bright and clear.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
GND	GND	Ground
VCC	5V	Power (3.3–5V modules)
SCL	A5	Clock
SDA	A4	Data

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Install the library: **Adafruit SSD1306 + Adafruit GFX Library** (Sketch > Include Library > Manage Libraries).
3. Tools > Board > Arduino Uno, then choose the Port.
4. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
5. Result: The screen shows 'IoT for Kids!'. If it stays blank, swap SDA and SCL or try address 0x3D in the code.

WHY IT IS USED

- Showing sensor readings in weather stations
- Smart watches and fitness bands
- Menus and messages on small devices
- Gadget status screens

In our workshop: Used in the Plant Helper and Weather Station so projects can display their own messages.

TowerPro SG90 Micro Servo Motor (180°)



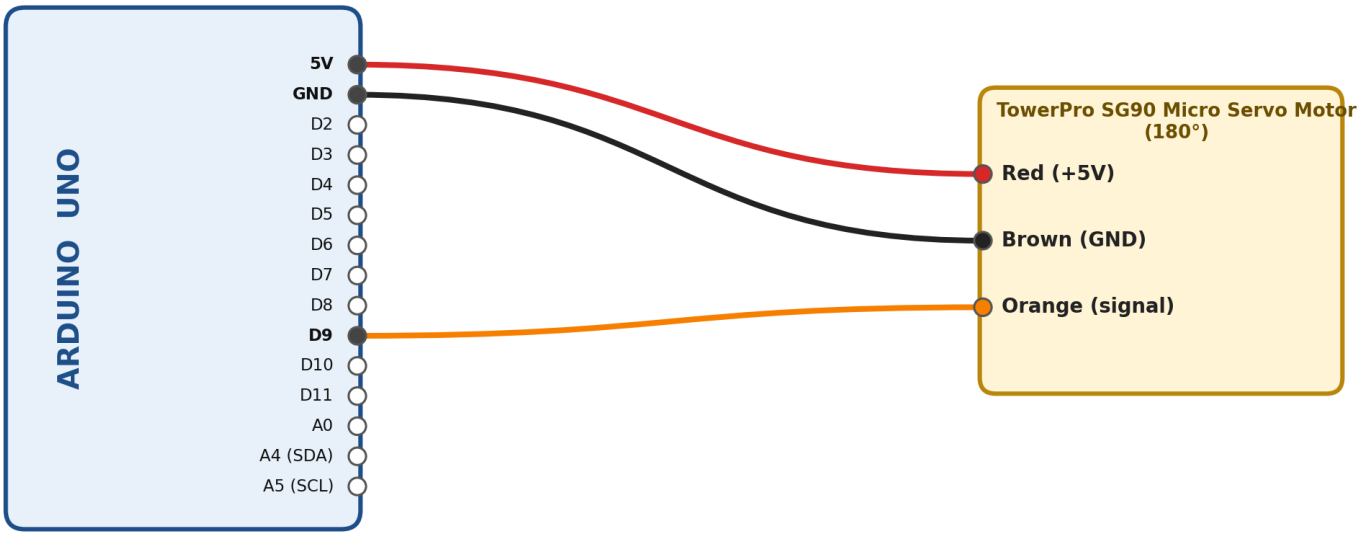
WHAT IT DOES

A small motor that turns to an exact angle from 0° to 180°. Used for robot arms, gates, doors and steering.

HOW IT WORKS

The Arduino sends short pulses. The length of each pulse tells the servo which angle to move to, and it holds that position. The Servo library does the timing for us.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Wire colour	Arduino pin	Purpose
Orange	D9	Signal
Red	5V	Power
Brown	GND	Ground

SETUP STEPS

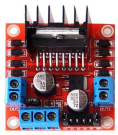
1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Result: The arm moves to 0°, 90° and 180°, and repeats. If it jitters, use a separate battery pack and join the grounds.

WHY IT IS USED

- Robot arms and grippers
- Automatic doors, gates and dustbin lids
- Camera and sensor pan/tilt
- Steering in model cars and RC planes

In our workshop: Moves things to a precise angle: the Smart Dustbin uses it to open and close the lid.

Motor Driver L298N + DC Motor





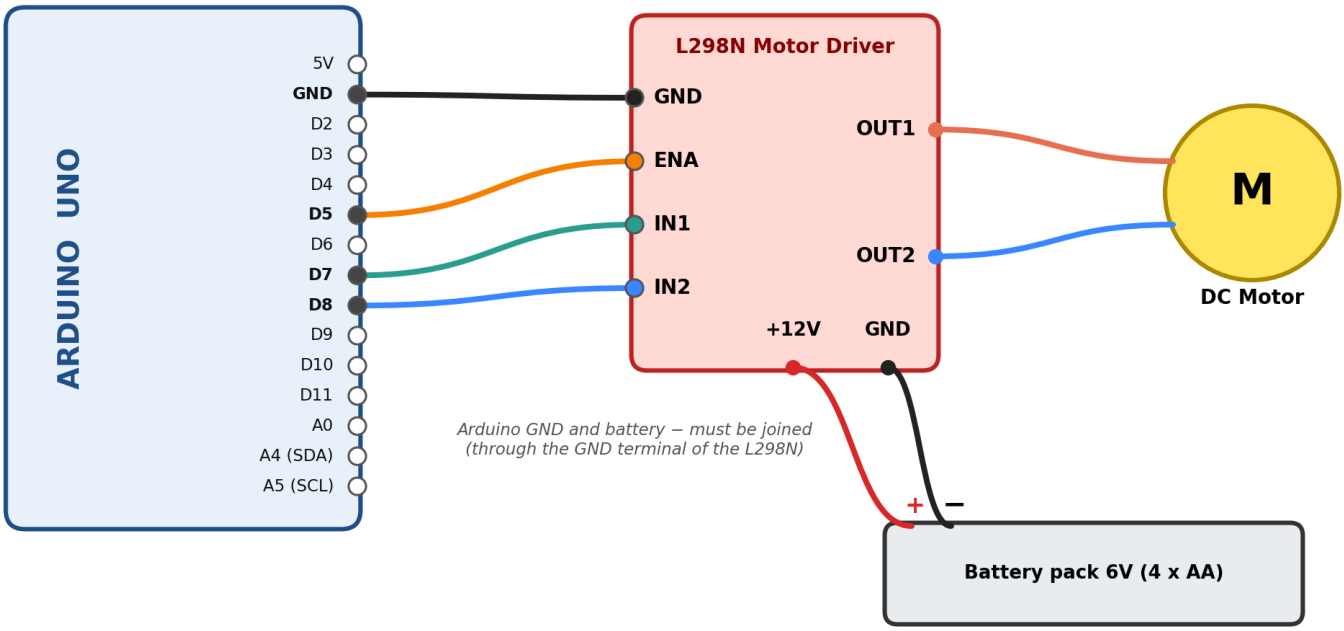
WHAT IT DOES

The L298N driver lets the Arduino control the speed and direction of DC motors. Used in robot cars, fans and moving models.

HOW IT WORKS

The Arduino pins are too weak to power a motor. The L298N works like a switch: the Arduino sends tiny signals (IN1, IN2 = direction, ENA = speed) and the driver sends the battery's power to the motor.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Connection	Goes to	Purpose
ENA	D5 (~)	Speed (PWM)
IN1 / IN2	D7 / D8	Direction
GND	Arduino GND	Common ground
OUT1 / OUT2	Motor wires	Motor power
+12V / GND	Battery + / −	Motor power supply

SETUP STEPS

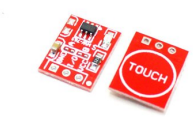
1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Remove the small jumper cap on the ENA pins so speed control works.
3. Connect the motor to OUT1 and OUT2, and the battery pack to +12V and GND on the driver. Connect the battery last, after the teacher checks it.
4. Tools > Board > Arduino Uno, then choose the Port.
5. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
6. Result: The motor turns forward for 2 seconds, stops, then turns backward for 2 seconds.

WHY IT IS USED

- Robot cars and rovers
- Fans and pumps
- Conveyor belts and moving models
- Any project that needs spinning motion

In our workshop: Shows how a small Arduino can control a bigger motor safely: the Fan Controller project uses it.

TTP223 Touch Switch Module



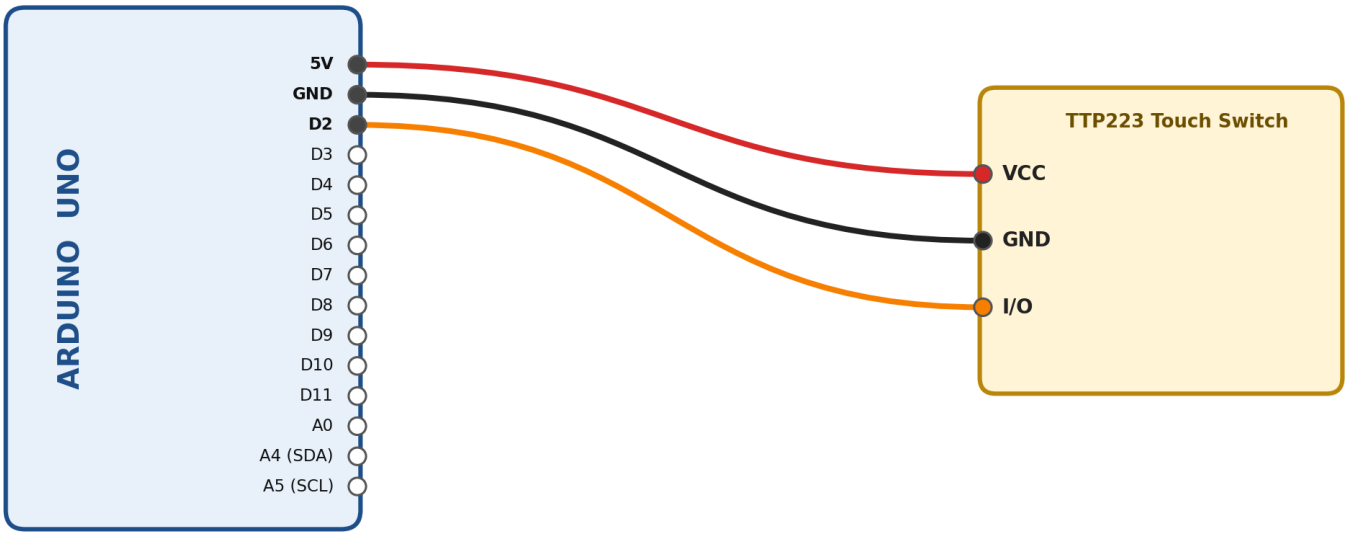
WHAT IT DOES

A switch that works when you touch the round pad. Used for touch lamps, switches and quiz buttons.

HOW IT WORKS

The chip senses the tiny electrical change when a finger comes close to the pad (capacitive sensing, like a phone screen). It then sends HIGH (5V) to the Arduino.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
VCC	5V	Power
GND	GND	Ground
I/O	D2	Output: HIGH when touched

SETUP STEPS

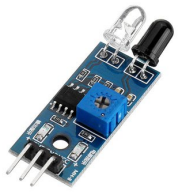
1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Result: The board's LED lights while you touch the pad.

WHY IT IS USED

- Touch lamps and touch-panel buttons
- Quiz and game buttons
- Appliance controls with no moving parts
- Hidden or waterproof switches

In our workshop: A friendly first input: students touch the pad and see an immediate reaction.

IR Obstacle Avoidance Sensor Module



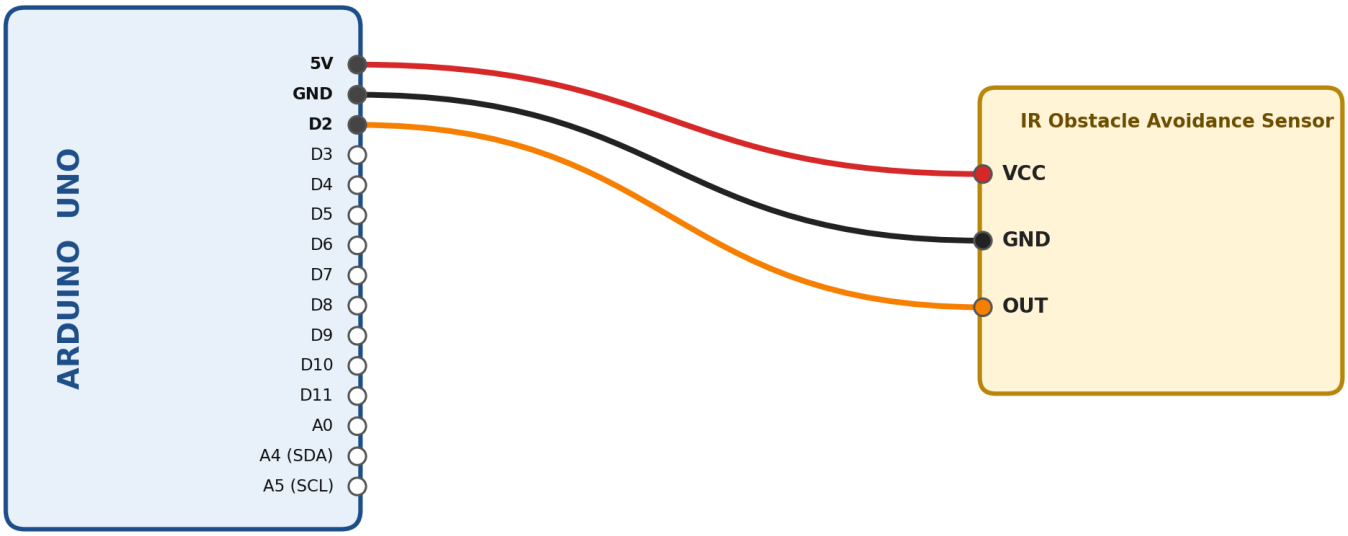
WHAT IT DOES

Detects an object in front of it, from a few centimetres up to about 30 cm. Used in obstacle-avoiding robots, counters and automatic taps.

HOW IT WORKS

One clear LED sends out invisible infrared light. If an object is in front, the light bounces back to the black receiver. The blue screw adjusts the detection distance, and a small LED on the board lights up when it sees something.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
VCC	5V	Power
GND	GND	Ground
OUT	D2	LOW when an object is detected

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Open Tools > Serial Monitor and set 9600 baud.
5. Result: 'Obstacle!' appears when a hand is in front of the sensor. Turn the blue screw to change the distance.

WHY IT IS USED

- Obstacle-avoiding robots
- Automatic taps, hand dryers and doors
- Object and people counters
- Line-following robots

In our workshop: Teaches how a machine can 'see' an object in front of it without touching it.

HC-SR04 Ultrasonic Distance Sensor



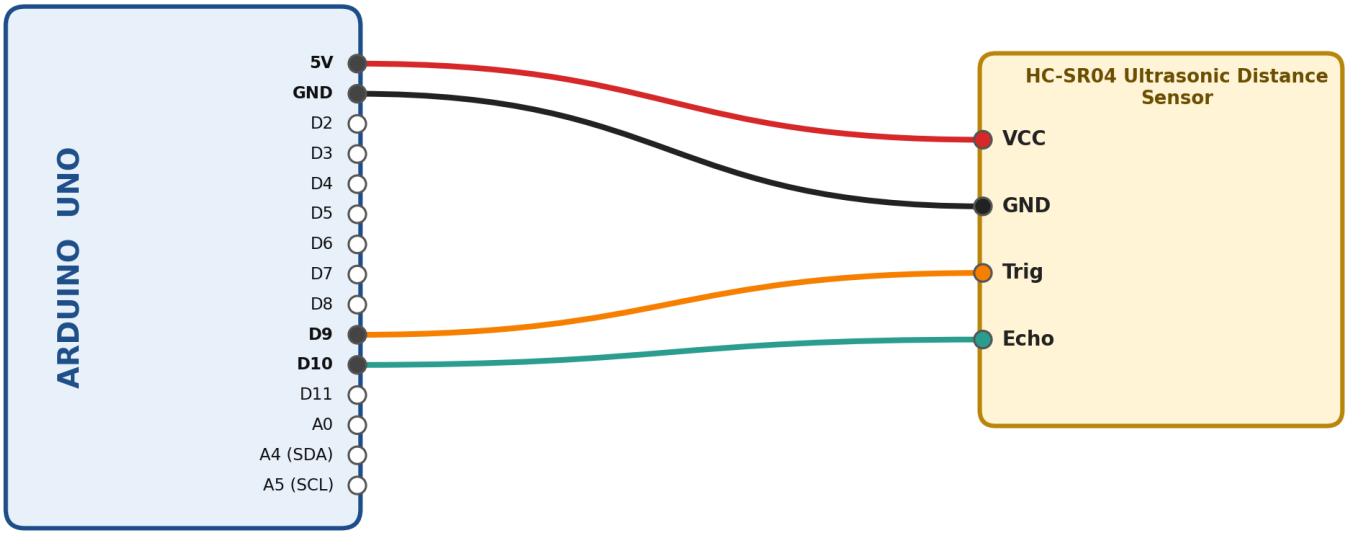
WHAT IT DOES

Measures distance to an object from about 2 cm to 4 m. Used for parking sensors, robots and water-level checks.

HOW IT WORKS

One side (T) sends a short, high-pitched sound that people can't hear. The other side (R) listens for the echo. The longer the echo takes, the farther the object is, just like bats and dolphins.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
VCC	5V	Power
Trig	D9	Start the sound pulse
Echo	D10	Receive the echo time
GND	GND	Ground

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Open Tools > Serial Monitor and set 9600 baud.
5. Result: The distance in cm is printed every 0.3 seconds.

WHY IT IS USED

- Car parking sensors
- Robots that avoid walls
- Water-tank level monitoring
- Automatic dustbins and hand-free devices

In our workshop: The 'magic ruler' of Day 1 and the trigger for the Smart Dustbin project.

Heat Flame Sensor



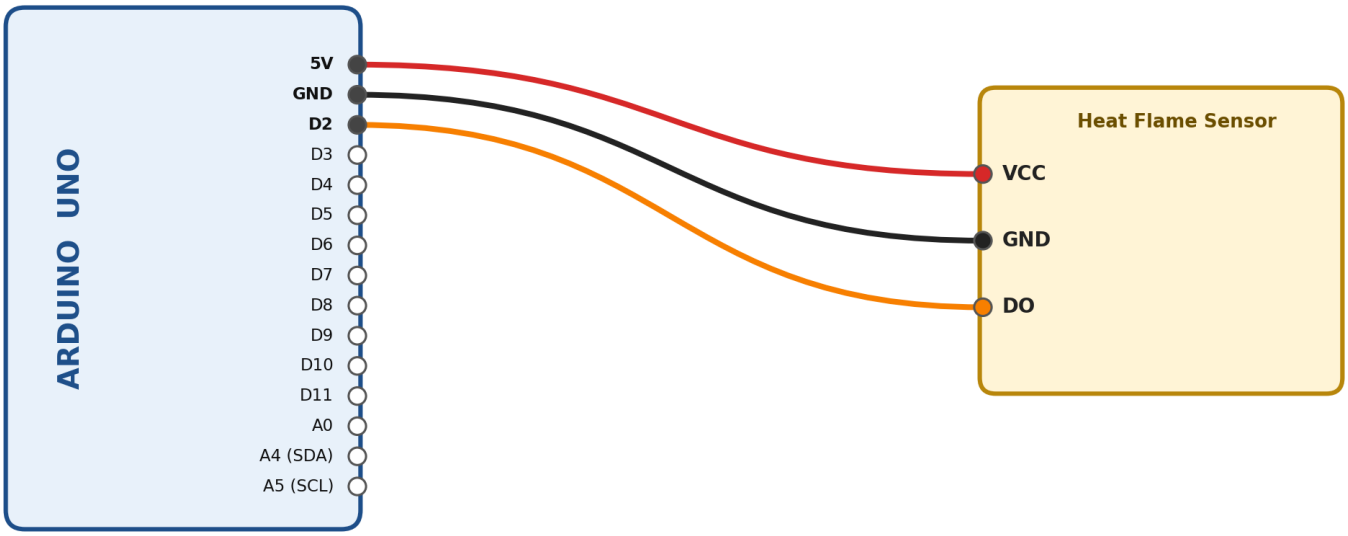
WHAT IT DOES

Detects the infrared light given off by a flame. Used in fire alarms and safety projects.

HOW IT WORKS

The black LED-shaped part is an infrared receiver. Fire gives out a lot of infrared light, so the sensor's output changes when a flame is near. The blue screw sets how sensitive it is.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
VCC	5V	Power
GND	GND	Ground
DO	D2	Digital output (usually LOW when flame seen)

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Optional: connect a buzzer (S to D8, − to GND) for the alarm sound.
3. Tools > Board > Arduino Uno, then choose the Port.
4. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
5. Open Tools > Serial Monitor and set 9600 baud.
6. Result: 'FIRE ALERT!' appears and the buzzer sounds when a flame is seen. Only the teacher runs the demo.

WHY IT IS USED

- Fire alarms in homes, schools and factories
- Gas stoves and boiler safety
- Kitchen and lab safety systems
- Forest and warehouse fire detection

In our workshop: Used in the teacher-run Fire Alarm demo to show how sensors protect people.

Raindrop Detection Sensor Module



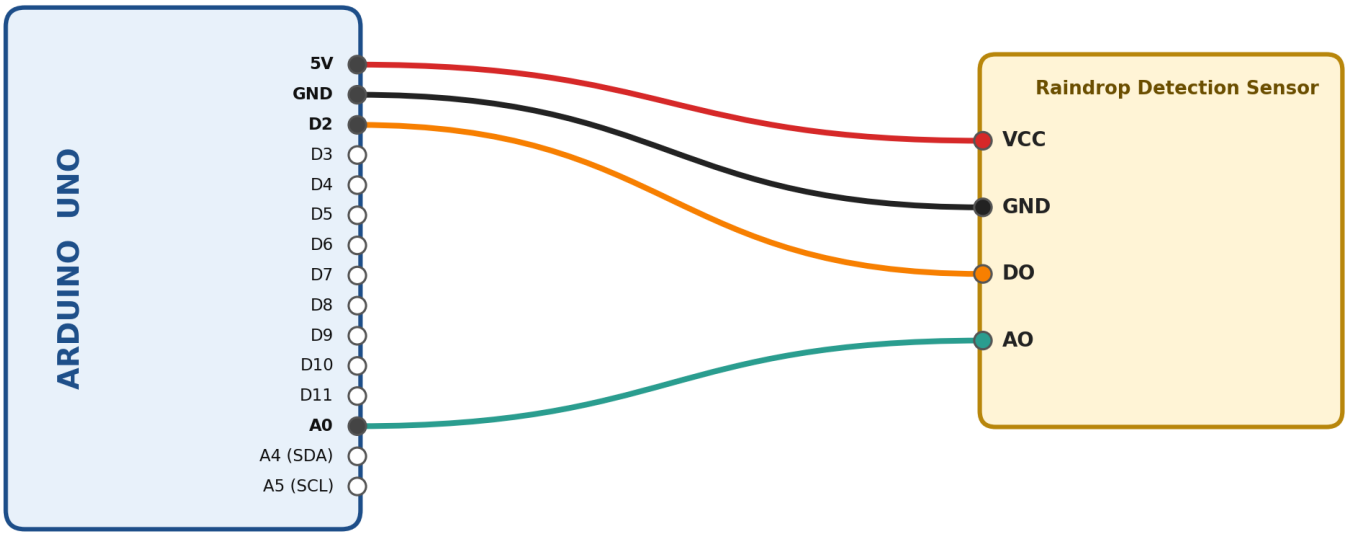
WHAT IT DOES

Senses water drops on a plate. Used for rain alarms, automatic wipers and cloth/roof covers.

HOW IT WORKS

The plate has thin metal tracks. When water falls on them it joins the tracks together and electricity can flow, so the reading goes down. The small board turns this into a number on AO and an ON/OFF signal on DO.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
VCC	5V	Power
GND	GND	Ground
AO	A0	Analog value (0–1023)
DO	D2	Digital output (optional)

SETUP STEPS

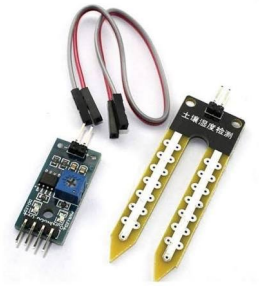
1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Connect the plate to the small board with the 2-wire cable. Optional: buzzer S to D8, − to GND.
3. Tools > Board > Arduino Uno, then choose the Port.
4. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
5. Open Tools > Serial Monitor and set 9600 baud.
6. Result: The numbers drop when drops land on the plate. Below 500 the buzzer sounds.

WHY IT IS USED

- Automatic windscreen wipers in cars
- Rain alarms for drying clothes
- Smart roof covers and window closers
- Weather stations and irrigation control

In our workshop: Used in the Rain Alert project: when drops hit the plate, a buzzer and light warn us.

Soil Moisture Sensor Module



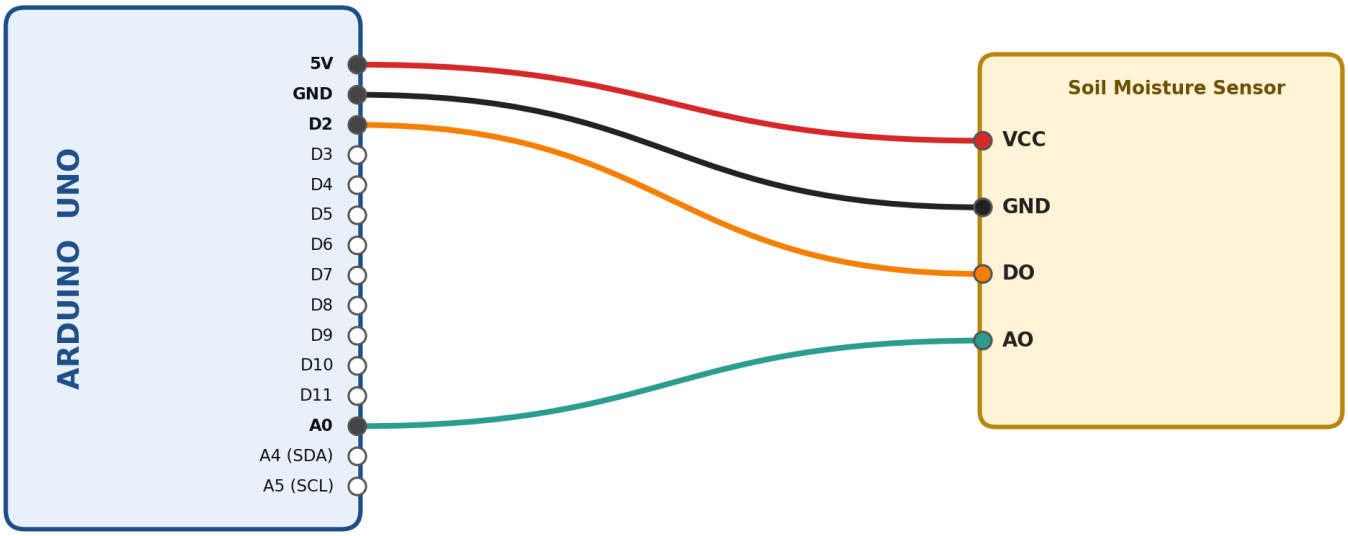
WHAT IT DOES

Measures how wet or dry soil is. Used for plant-watering reminders and smart irrigation.

HOW IT WORKS

Wet soil conducts electricity better than dry soil. The two prongs measure this and the small board turns it into a number: the exact values differ per sensor, so the class calibrates it.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
VCC	5V	Power
GND	GND	Ground
AO	A0	Analog value (dry = high)
DO	D2	Digital output (optional)

SETUP STEPS

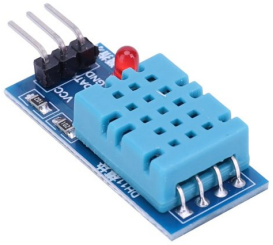
1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Tools > Board > Arduino Uno, then choose the Port.
3. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
4. Open Tools > Serial Monitor and set 9600 baud.
5. Result: Moisture numbers appear. 'Plant needs water!' appears when the soil is dry (calibrate 700).

WHY IT IS USED

- Automatic plant watering systems
- Smart farming and irrigation
- Gardening reminders
- Greenhouse monitoring

In our workshop: Used in the Plant Helper project to tell students when a plant needs water.

DHT11 Humidity & Temperature Sensor



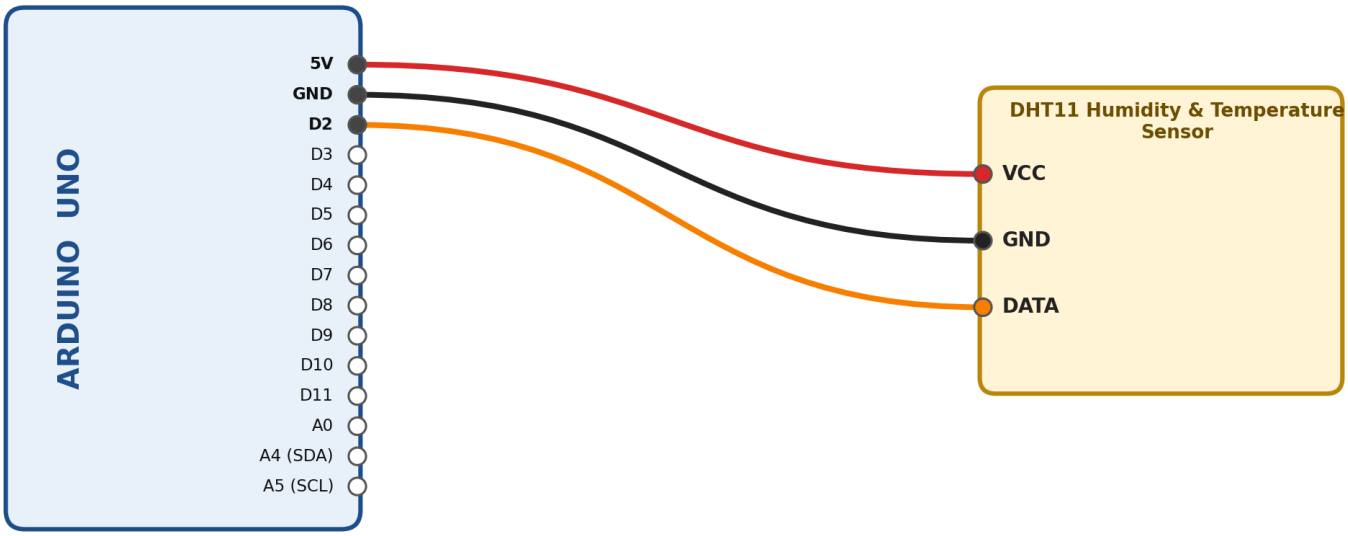
WHAT IT DOES

Measures the temperature and the humidity of the air. Used in weather stations, smart rooms and greenhouses.

HOW IT WORKS

It contains a humidity sensor and a temperature sensor. It sends the numbers to the Arduino as a digital message on a single DATA wire, about once every two seconds.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
VCC (+)	5V	Power
DATA (S)	D2	Digital data
GND (−)	GND	Ground

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Install the library: **DHT sensor library (Adafruit) + Adafruit Unified Sensor** (Sketch > Include Library > Manage Libraries).
3. Tools > Board > Arduino Uno, then choose the Port.
4. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
5. Open Tools > Serial Monitor and set 9600 baud.
6. Result: Temperature and humidity are printed every 2 seconds.

WHY IT IS USED

- Weather stations
- Smart rooms and air-conditioning control
- Greenhouses and cold storage
- Baby rooms and storage conditions

In our workshop: Measures temperature and humidity for the Mini Weather Station and the sensor stations on Day 2.

DS18B20 Digital Temperature Sensor



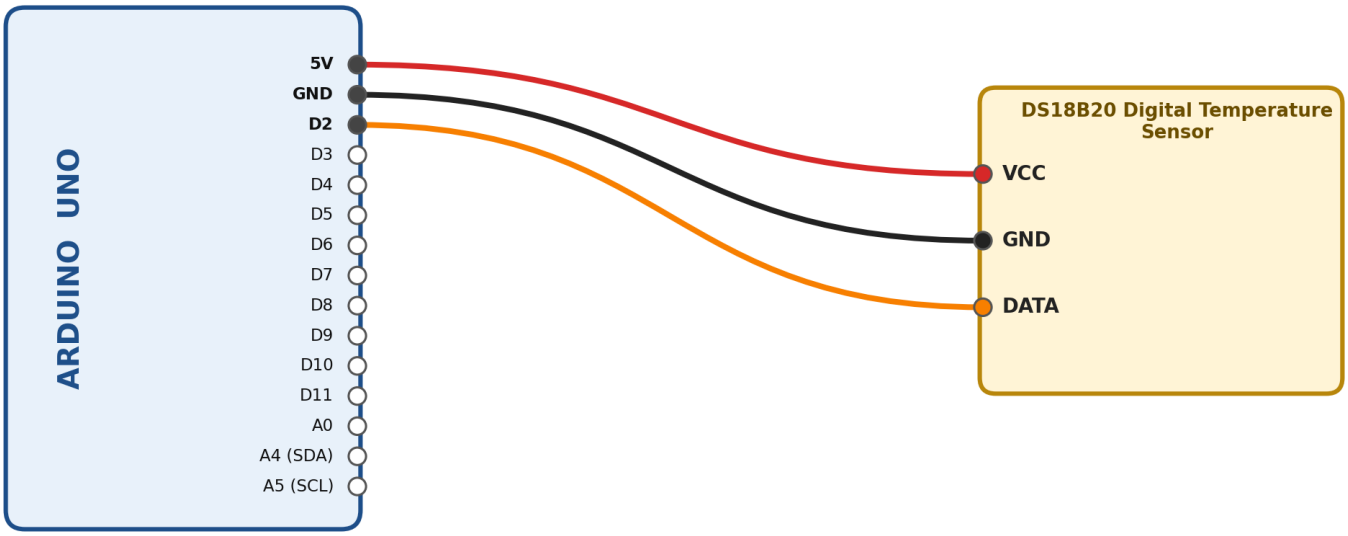
WHAT IT DOES

A more accurate digital thermometer. Used in thermometers, water-temperature checks and temperature control.

HOW IT WORKS

Each DS18B20 sensor has its own unique number and talks to the Arduino on one wire (the 1-Wire protocol), sending the temperature as digital data. This module already includes the pull-up resistor it needs.

CIRCUIT DIAGRAM: connect the Arduino to the module as shown



WIRING TABLE

Module pin	Arduino pin	Purpose
VCC (+)	5V	Power
DATA (S)	D2	1-Wire data
GND (−)	GND	Ground

SETUP STEPS

1. Unplug the USB cable. Wire the parts exactly as in the diagram and table.
2. Install the library: **OneWire and DallasTemperature** (Sketch > Include Library > Manage Libraries).
3. Tools > Board > Arduino Uno, then choose the Port.
4. Open this component's sketch from the separate Code PDF (or Arduino_Sketches.zip) and click Upload.
5. Open Tools > Serial Monitor and set 9600 baud.
6. Result: The temperature in °C is printed every second. -127 means a loose DATA wire.

WHY IT IS USED

- Accurate digital thermometers
- Water and liquid temperature checks
- Aquariums and fridges
- Temperature-controlled devices such as fans and heaters

In our workshop: Used to compare with the DHT11 and to switch the fan on in the Fan Controller project.

PART C · Project guides for Day 2

Wire each project with its table. 'Rail' means the breadboard 5V (red) or GND (blue) rail shared with the Arduino. The sketches are in the separate Code PDF.

Project 1 · Smart Dustbin

The lid opens when a hand comes close.

Sense: a hand comes within 15 cm (HC-SR04)
Think: the Arduino checks if distance is less than 15
Act: the servo opens the lid for 3 seconds, then closes it

LIBRARIES

None (Servo is built in)

Part and pin	Connects to
HC-SR04 VCC / GND	5V rail / GND rail
HC-SR04 Trig	D9
HC-SR04 Echo	D10
Servo orange (signal)	D6
Servo red / brown	5V rail / GND rail

Project 2 · Rain Alert

A buzzer and a red light warn when rain is detected.

Sense: rain drops on the plate (rain sensor)
Think: the Arduino checks if the value is below the limit
Act: the buzzer sounds and the red light turns on

LIBRARIES

None

Part and pin	Connects to
Rain board VCC / GND	5V rail / GND rail
Rain board AO	A0
Buzzer S	D7
Buzzer −	GND rail
Traffic light R	D8
Traffic light GND	GND rail

Project 3 · Plant Helper

Shows if the plant needs water: red = dry, green = happy, with a message on the OLED.

Sense: soil wetness (soil moisture sensor)
Think: the Arduino checks if the soil is dry
Act: the LED turns red or green and the OLED shows a message

LIBRARIES

Adafruit SSD1306, Adafruit GFX Library

Part and pin	Connects to
Soil board VCC / GND	5V rail / GND rail
Soil board AO	A0
RGB R / G / B	D9 / D10 / D11
RGB GND	GND rail
OLED VCC / GND	5V rail / GND rail
OLED SDA / SCL	A4 / A5

Project 4 · Fire Alarm (teacher-run test)

Sounds an alarm and lights the red lamp when a flame is detected.

Sense: a flame (flame sensor, teacher-run test)
Think: the Arduino checks if a flame is detected
Act: the buzzer sounds and the red light turns on

LIBRARIES

None

Part and pin	Connects to
Flame sensor VCC / GND	5V rail / GND rail
Flame sensor DO	D2
Buzzer S	D7
Buzzer −	GND rail
Traffic light R	D8
Traffic light GND	GND rail

Project 5 · Mini Weather Station

Shows temperature and humidity on the OLED screen.

Sense: temperature and humidity (DHT11)
Think: the Arduino reads the sensor every 2 seconds
Act: the OLED shows the temperature and humidity

LIBRARIES

DHT sensor library, Adafruit Unified Sensor, Adafruit SSD1306, Adafruit GFX Library

Part and pin	Connects to
DHT11 VCC / GND	5V rail / GND rail
DHT11 DATA	D2
OLED VCC / GND	5V rail / GND rail
OLED SDA / SCL	A4 / A5

Project 6 · Fan Controller (stretch)

The motor starts when the temperature goes above a set value.

Sense: temperature (DS18B20)
Think: the Arduino checks if it is above 30 °C
Act: the motor starts and the fan spins

LIBRARIES

OneWire, DallasTemperature

Part and pin	Connects to
DS18B20 VCC / GND	5V rail / GND rail
DS18B20 DATA	D2
L298N ENA / IN1 / IN2	D5 / D7 / D8
L298N GND	Arduino GND
Motor	OUT1 / OUT2
Battery + / −	+12V / GND on driver